# AFOLABI OLAWALE GOODLUCK

**Machine Learning Engineer · Data Scientist · AI Systems Builder**

Lagos, Nigeria | lekanolawale477@gmail.com | +234 808 441 5996 | LinkedIn | GitHub | Portfolio

## PROFESSIONAL SUMMARY

ML Engineer and Data Scientist with hands-on experience building production predictive models, NLP pipelines, and AI systems end to end. Strong mathematical foundation (B.Sc. Mathematics) applied to real-world problems — from e-sign propensity modelling in digital lending to fraud-risk classification and multi-agent AI for fintech. Comfortable across the full ML lifecycle: data cleaning, feature engineering, model training and evaluation, and cloud deployment on AWS, including model-routing and unit-economics decisions that keep AI systems cost-effective at production scale.

## SELECTED IMPACT

- **RubiesAI:** built the agentic banking assistant that lets Rubies Bank customers check balances, buy, and transfer money by chat alone — Qucoon's flagship agentic-banking build.
- **1,000+ daily queries** routed to 5 specialised agents at 89% first-contact resolution, with 85% of conversations handled fully autonomously (Motito).
- **70% faster documentation** (GeroCare two-pass validated SOAP generation) and $R^2 = 0.87$ on held-out salary prediction data.

## TECHNICAL SKILLS

**ML / AI:** scikit-learn · TensorFlow · CNN · NLP · RAG · LLM Integration · Prompt Engineering · Agentic AI · Model Routing & Cost Optimisation

**Data & Stats:** Pandas · NumPy · Feature Engineering · EDA · Regression · Classification · Clustering · Model Evaluation (AUC-ROC, Precision/Recall, F1)

**NLP:** NLTK · TF-IDF · Sentiment Analysis · Text Classification · Semantic & Hybrid Search · Embeddings (Titan / OpenAI)

**Databases:** PostgreSQL + pgvector · Pinecone · Redis · MySQL · Supabase · DynamoDB · SQL

**Viz & BI:** Power BI · Plotly · Matplotlib · Seaborn · Excel · Streamlit

**Cloud & Dev:** AWS (Lambda, S3, Bedrock, ECS Fargate, IAM, VPC, CloudWatch, SQS/SNS) · Docker · FastAPI · GitHub Actions · Python

## WORK EXPERIENCE

### Qucoon Limited — AI / ML Engineer

*Mar 2025 – Present*

*Lagos, Nigeria · Client-facing model delivery, internal AI product development, and cloud architecture*

**RubiesAI — Agentic Banking Assistant** — *Rubies Bank (Qucoon Client Engagement) · AWS Bedrock · Agentic Tool-Use · Production — Flagship Project*

- Built an agentic AI banking assistant for Rubies Bank that performs real banking actions — **checking balances, buying, and transferring money** — directly from a customer's natural-language chat message.
- Designed a multi-tool agentic architecture on AWS Bedrock (Claude) where the model selects and calls the right banking tool per user intent; a backend execution layer validates and carries out the action against the bank's core systems.
- Built a dual-intent router separating genuine transactional requests from ordinary conversation, with per-transaction business-rule checks and confirmation before any money movement executes.

**Motito — Multi-Agent Customer Support System** — *NLP · Intent Classification · RAG · Production*

- Built an intent-classification system using AWS Bedrock LLMs to route **1,000+ daily queries** to 5 specialised agents, achieving **89% first-contact resolution**.
- Designed a dual-store retrieval pipeline (PostgreSQL + pgvector, Titan 512-dim embeddings, IVFFlat cosine index) for both product search and the support knowledge base, using Claude 3.5 Sonnet to turn raw product JSON into search-oriented descriptions before embedding.
- Built a fully async, serverless ETL pipeline (API → SQS → Lambda) that decouples slow LLM/embedding calls from the request path, with incremental per-portal Chatwoot sync and idempotent upserts — fully automated so the knowledge base stays current with no manual intervention.
- Implemented confidence scoring (0.0–1.0) to auto-escalate low-certainty responses, keeping AI handling rate at **85%**.

**FlowOS — Model Routing & AI Unit Economics** — *AWS Bedrock · Cost Modelling · Hybrid Search*

- Verified real AI cost-per-reply unit economics across a multi-model routing strategy (Bedrock Claude Haiku 4.5 primary, Nova Pro fallback, Titan Embed Text v2), keeping a multi-tenant WhatsApp AI platform commercially sustainable at scale.

- Built a hybrid semantic + keyword product-search layer (pgvector similarity blended with full-text ranking) so near-miss and plural/keyword queries are never missed by embedding similarity alone.

**VestAI — AI Disclosure & News Summarisation Pipelines** — *Qoovest · AWS Bedrock · Serverless ETL*

- Built a Bedrock-Claude summarisation step inside a serverless disclosure pipeline, turning raw NGX financial filings (PDF) into plain-English, retail-investor-ready summaries with a human-in-the-loop quality gate before publishing.
- Built a market-news pipeline that fetches and deduplicates stories from multiple Nigerian finance outlets, groups same-event coverage, rates importance with Bedrock, and serves the right stories to the right investors through a dedicated news API.

**GeroCare — AI Medical Copilot** — *Healthcare NLP · AWS · Real-Time Transcription*

- Built a real-time voice-to-text transcription pipeline (Amazon Transcribe) and SOAP report generation with Claude Sonnet 4, using Amazon Nova Pro for fast real-time classification — cutting documentation time by **70%**.
- Added a two-pass validation model: first pass generates the draft, second pass cross-checks every claim against the raw transcript and flags unsupported outputs.
- Designed a read-only DB architecture so the model reads patient history from MySQL but only writes drafts to S3 — zero risk to live clinical records.

**Qorpy — Enterprise RAG FAQ System** — *Vector Search · Multi-Tenant · Serverless*

- Shipped a multi-tenant semantic search platform using Pinecone + Titan embeddings, with sub-query decomposition for **3× faster retrieval** (1.75s median latency).
- Built and deployed the full ML inference pipeline on AWS Lambda with GitHub Actions CI/CD — updates ship to production in under 2 minutes.

## PROJECTS

- **Digital Lending E-Sign Predictor** (scikit-learn · Classification · Feature Engineering, 2024): trained a classifier to predict e-sign completion, enabling fintech teams to focus outreach on high-drop-off segments.
- **Sentiment Lab — Production NLP Pipeline** (NLTK · TF-IDF · Naive Bayes, 2024): modular, API-ready sentiment pipeline benchmarking TF-IDF vs. modern embedding approaches across multiple classifiers.
- **Sign Language Recognition — CNN** (TensorFlow · CNN · Computer Vision, 2024): trained a CNN to classify hand gestures for real-time sign-language translation.
- **Nigerian RAG Health System** (RAG · Embeddings · FastAPI, 2024): RAG-powered health Q&A system localised for Nigerian healthcare context.
- **Smart Salary Predictor** (scikit-learn · Regression · EDA, 2024): end-to-end pipeline achieving $R^2 = 0.87$ on held-out data; identified education, experience, and role as top salary drivers.

## EDUCATION

**Lagos State University** *Aug 2021 – Jan 2025*

*B.Sc. Mathematics & Education*